

# Building Maps on the Web Using RStudio, Leaflet, & Shiny

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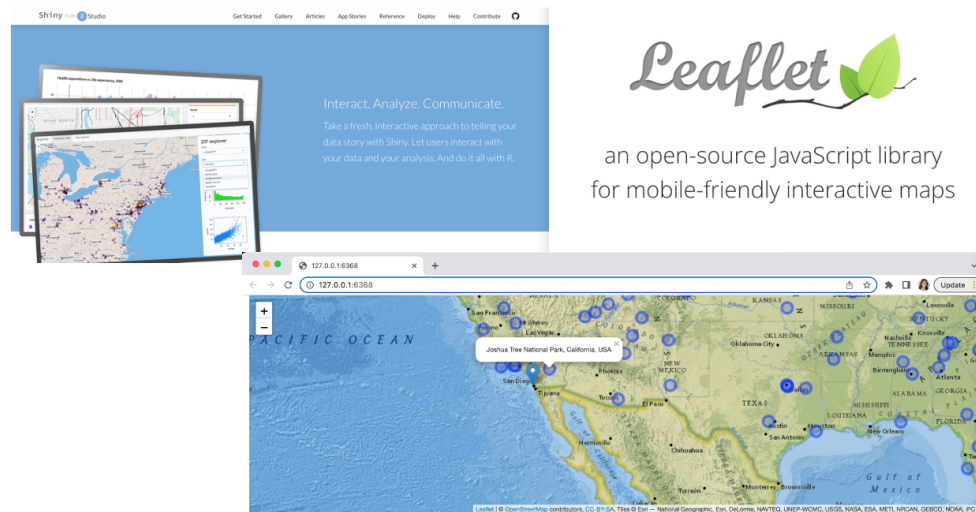


Figure 1: Shiny, Leaflet and RStudio will be used to create maps and annotate them with spatial data.

## ABSTRACT

This lab will provide a hands-on introduction to visualization of spatial data using interactive maps that can be deployed as public web pages. We will use a combination of RStudio, the Shiny package, and the Leaflet open-source library to provide an introduction on how to combine data and maps to create public web pages. Attendees will gain an overview of RStudio, Leaflet, and Shiny Applications. They will learn how to install packages for leaflet and Shiny, create and customize different types of leaflet maps, including a choropleth, and develop a Shiny application deployable on the web.

## CCS CONCEPTS

• **Computer systems organization** → **Embedded systems**; *Redundancy*; Robotics; • **Networks** → Network reliability.

## KEYWORDS

RStudio, Shiny, Leaflet, Data Visualization, Spatial Mapping

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## 1 INTRODUCTION

Maps provide an intuitive interface while communicating elements that are related spatially and presented visually. Any data holding a spatial component lends itself to presentation on a map. Interactive maps allow us to explore spatial data, build layers, identify patterns, drill down to reveal additional information, and inform data-driven decisions. This lab will introduce attendees to the seamless integration between RStudio, Shiny, and the Leaflet Mapping Library to enable the creation of spatially mapped data on the web with minimal friction.

## 2 OBJECTIVES

The overall objective of this lab is to introduce the mapping of spatial data using RStudio and create web visualizations containing mapped data using Shiny.

After completing this hands on lab, attendees will be able to:

- Describe key features of RStudio, Leaflet, and Shiny
- Download and install each package.
- Create Base Maps

- Annotate base maps
- Including adding Markers, Circles, and Polygons
- Add and format Popups and Labels elements
- Create a Choropleth Maps
- Integrate Shiny and Leaflet to display maps on the web

In summary, upon completing this hands on lab, attendees will be equipped with enough knowledge to create various maps and populate the maps with makers, popups, and other interactive elements. We will use real data to inform the appearance of a choropleth

map and finally deploy the application to the web seamlessly using Shiny.

### 3 INTENDED AUDIENCE

This lab is suitable for attendees that would like to learn more about mapping data and using the Shiny app with RStudio to deploy the maps to the web. Attendees should have some programming experience, but not necessarily in RStudio.